

Brown and Brook Trout Survival Analysis

STAT 3559 Project Analysis Journal

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I. Introduction

Brook and Brown Trout are an ecologically and economically important species in Pennsylvania, as they serve as key indicators of stream health and support the recreational fishing industries across the state. Programs such as stocking have been used to supplement and manage fisheries across the state. Understanding what drives survival time in stocked trout is essential for informing effective and ethical conservation and management strategies.

Our project examines the survival of Brown (*Salmo trutta*) and Brook (*Salvelinus fontinalis*) Trout across five Pennsylvania streams over a 153-day observation period. Fish were tagged and monitored for survival and movement behavior. Our primary research question is: Which fish-level and stream-level characteristics are associated with trout survival time?

We used Kaplan-Meier survival estimates and log-rank tests to compare survival between groups. This was followed by Cox Proportional Hazards regression to identify which explanatory variables predict time-to-death.

II. Data Summary

Data Source

Data were collected as part of a field study tracking radio-tagged trout in five streams in Pennsylvania. Each fish was implanted with a radio transmitter and monitored at regular intervals; the last confirmed observation date and survival status were recorded. The dataset represents a prospective cohort study of 99 individual fish.

General Metadata:

Feature	Detail
Who	Individual Brook and Brown Trout
What	Survival time and associated fish/stream characteristics
When	Observation window of 153 days
Where	Five streams in Pennsylvania, USA
Why	Understand factors influencing salmonid survival in small streams

Variable Descriptions

Variable	Type	Description
<code>fish_id</code>	Identifier	Unique numeric ID for each fish (excluded from analysis)
<code>species</code>	Categorical	Fish species: Brook or Brown Trout
<code>stream</code>	Categorical	Stream where fish was stocked (5 streams)
<code>order</code>	Ordinal	Stream order (ecological complexity): low (1), medium (2), complex (4)
<code>Lgt</code>	Continuous	Fish length (mm) at time of stocking
<code>Wgt</code>	Continuous	Fish weight (grams) at time of stocking
<code>cumdrain</code>	Continuous	Cumulative watershed drainage area — higher values indicate a larger catchment
<code>move_rate</code>	Continuous	Mean daily movement rate (m/day), recorded via radio transmitter
<code>time2</code>	Continuous	Survival time: days until death or last observed alive (max 166 days)
<code>death2</code>	Binary (event)	Survival outcome: Yes = died or lost, No = still alive (censored)

Variable Transformations:

- `order` recoded as an ordered factor: 1 → “low”, 2 → “medium”, 4 → “complex”
- `death2` recoded as factor: 0 → “No”, 1 → “Yes”; a parallel numeric `event` column (0/1) was created for use in `Surv()`
- `Wgt` and `move_rate` are log-transformed prior to Cox regression due to strong right skew
- `time1` and `death` (first-interval observations) were excluded; analysis focuses on the full follow-up period captured by `time2` and `death2`
- `Lgt` is excluded from Cox regression due to high multicollinearity with `Wgt` ($r = 0.90$)
- `stream` is excluded from the final Cox model because it is perfectly collinear with `order` (each stream maps to exactly one order value), which caused model instability

Data Loading & Preprocessing

```
library(tidyr)
library(dplyr)
library(ggplot2)
library(survival)
library(survminer)
library(knitr)
library(kableExtra)
library(gridExtra)

setwd("~/Downloads")
trout_survival <- read.csv("troutsurvival_daily.csv", header = TRUE)
```

```

trout_survival$order <- factor(trout_survival$order, levels = c(1, 2, 4),
                              labels = c("low", "medium", "complex"))
trout_survival$death2 <- factor(trout_survival$death2, levels = c(0, 1),
                               labels = c("No", "Yes"))
trout_survival$stream <- as.factor(trout_survival$stream)
trout_survival$species <- as.factor(trout_survival$species)

trout_survival <- trout_survival %>%
  select(-time1) %>%
  select(-death)

# Drop the 9 rows missing move_rate
trout_clean <- trout_survival %>% filter(!is.na(move_rate))

# Numeric event indicator (required for Surv())
trout_clean$event <- ifelse(trout_clean$death2 == "Yes", 1, 0)

# Log-transformed columns for regression
trout_clean$log_Wgt <- log(trout_clean$Wgt)
trout_clean$log_move_rate <- log(trout_clean$move_rate)

cat("Dimensions after removing NAs:", nrow(trout_clean), "observations,",
    ncol(trout_clean), "variables\n")

```

```
## Dimensions after removing NAs: 90 observations, 13 variables
```

Nine fish (9.1%) were excluded due to missing `move_rate` values, leaving **n = 90** complete observations for analysis.

Potential Issues:

- Missing movement data may reflect transmitter malfunction; if missingness is related to fish condition, estimates could be slightly biased.
- `stream` and `order` are perfectly collinear — each stream corresponds to exactly one order value — so both cannot be included in the same model. We retain `order` for its broader ecological interpretability.
- `time2` contains values exceeding 153 days (max = 166), suggesting some fish were tracked beyond the nominal study window; these observations are retained as valid.

Correlation Among Quantitative Predictors

`Lgt` and `Wgt` are highly correlated ($r = 0.90$), confirming both cannot be in the same model when we move onto predictive analysis. We retain `log_Wgt` and exclude `Lgt`. The remaining predictors show low pairwise correlations, so no additional collinearity concerns exist.

```

num_vars <- trout_clean[, c("Lgt", "Wgt", "move_rate", "cumdrain")]
cor(num_vars, use = "complete.obs")

```

```

##           Lgt           Wgt move_rate  cumdrain
## Lgt      1.0000000 0.9351954 0.2662183 0.3777878
## Wgt      0.9351954 1.0000000 0.2108855 0.3208412
## move_rate 0.2662183 0.2108855 1.0000000 0.2904264
## cumdrain  0.3777878 0.3208412 0.2904264 1.0000000

```

Exploratory Data Analysis

Response Variable: Survival Time (time2)

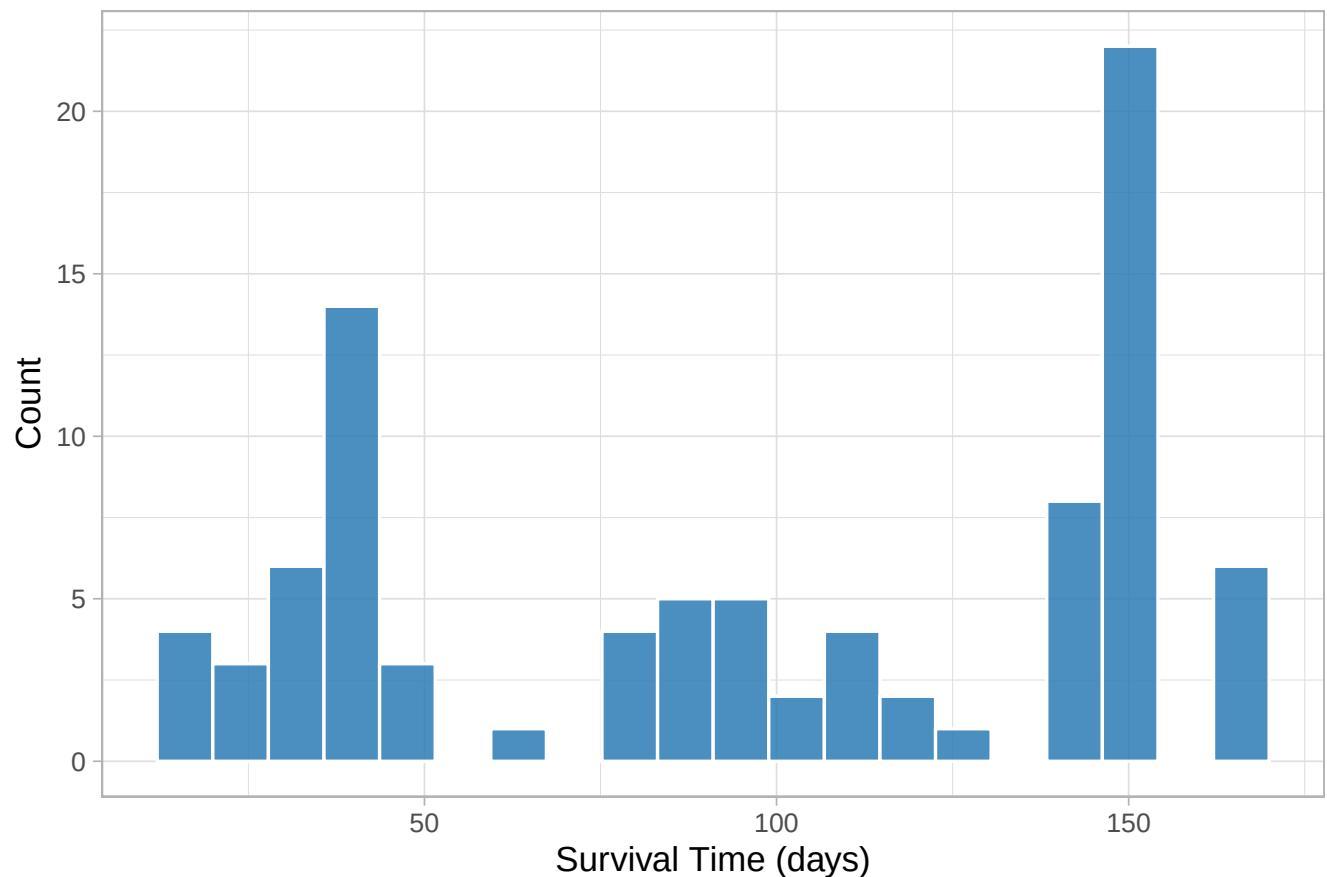
```
summary_stats <- trout_clean %>%  
  summarise(  
    Min    = min(time2),  
    Q1     = quantile(time2, 0.25),  
    Median = median(time2),  
    Mean   = round(mean(time2), 1),  
    Q3     = quantile(time2, 0.75),  
    Max    = max(time2),  
    SD     = round(sd(time2), 1)  
  )  
  
kable(summary_stats, caption = "Summary Statistics: Survival Time (days)") %>%  
  kable_styling(bootstrap_options = c("striped", "hover"), full_width = FALSE)
```

Table 1: Summary Statistics: Survival Time (days)

Min	Q1	Median	Mean	Q3	Max	SD
16	43	98	98.2	153	166	51.3

```
ggplot(trout_clean, aes(x = time2)) +  
  geom_histogram(bins = 20, fill = "#2c7bb6", color = "white", alpha = 0.85) +  
  labs(title = "Distribution of Survival Time",  
        x = "Survival Time (days)", y = "Count") +  
  theme_light(base_size = 13)
```

Distribution of Survival Time



The distribution of survival time is strongly left-skewed, with most fish surviving to or near the maximum follow-up period. This is expected given substantial right-censoring: 57 of 90 fish (63%) were still alive at last observation (`death2 = No`).

`time2` represents the number of days from stocking until either death or last confirmed observation, making it a classic **right-censored time-to-event outcome** directly suited to Kaplan-Meier estimation and Cox Proportional Hazards regression.

Categorical Variables

```
sp_tbl <- trout_clean %>% count(species) %>%  
  mutate(Proportion = round(n / sum(n), 3)) %>%  
  rename(Species = species, Count = n)  
  
st_tbl <- trout_clean %>% count(stream) %>%  
  mutate(Proportion = round(n / sum(n), 3)) %>%  
  rename(Stream = stream, Count = n)  
  
or_tbl <- trout_clean %>% count(order) %>%  
  mutate(Proportion = round(n / sum(n), 3)) %>%  
  rename(Order = order, Count = n)  
  
ev_tbl <- trout_clean %>% count(death2) %>%
```

```
mutate(Proportion = round(n / sum(n), 3)) %>%
  rename(Died = death2, Count = n)

kable(sp_tbl, caption = "Species Distribution") %>%
  kable_styling(bootstrap_options = "striped", full_width = FALSE)
```

Table 2: Species Distribution

Species	Count	Proportion
brook	47	0.522
brown	43	0.478

```
kable(st_tbl, caption = "Stream Distribution") %>%
  kable_styling(bootstrap_options = "striped", full_width = FALSE)
```

Table 3: Stream Distribution

Stream	Count	Proportion
Hunts_Run	29	0.322
McKinnon_Run	18	0.200
McNuff_Branch	18	0.200
Rock_Run	7	0.078
Whitehead_Run	18	0.200

```
kable(or_tbl, caption = "Stream Order Distribution") %>%
  kable_styling(bootstrap_options = "striped", full_width = FALSE)
```

Table 4: Stream Order Distribution

Order	Count	Proportion
low	7	0.078
medium	54	0.600
complex	29	0.322

```
kable(ev_tbl, caption = "Event (Death) Distribution") %>%
  kable_styling(bootstrap_options = "striped", full_width = FALSE)
```

Table 5: Event (Death) Distribution

Died	Count	Proportion
No	28	0.311
Yes	62	0.689

The sample is roughly balanced between Brook and Brown Trout and distributed across five streams. Stream order is dominated by medium-order streams (63%). Of the 90 fish in the cleaned dataset, 62 (68.9%) experienced the event (death or loss).

```

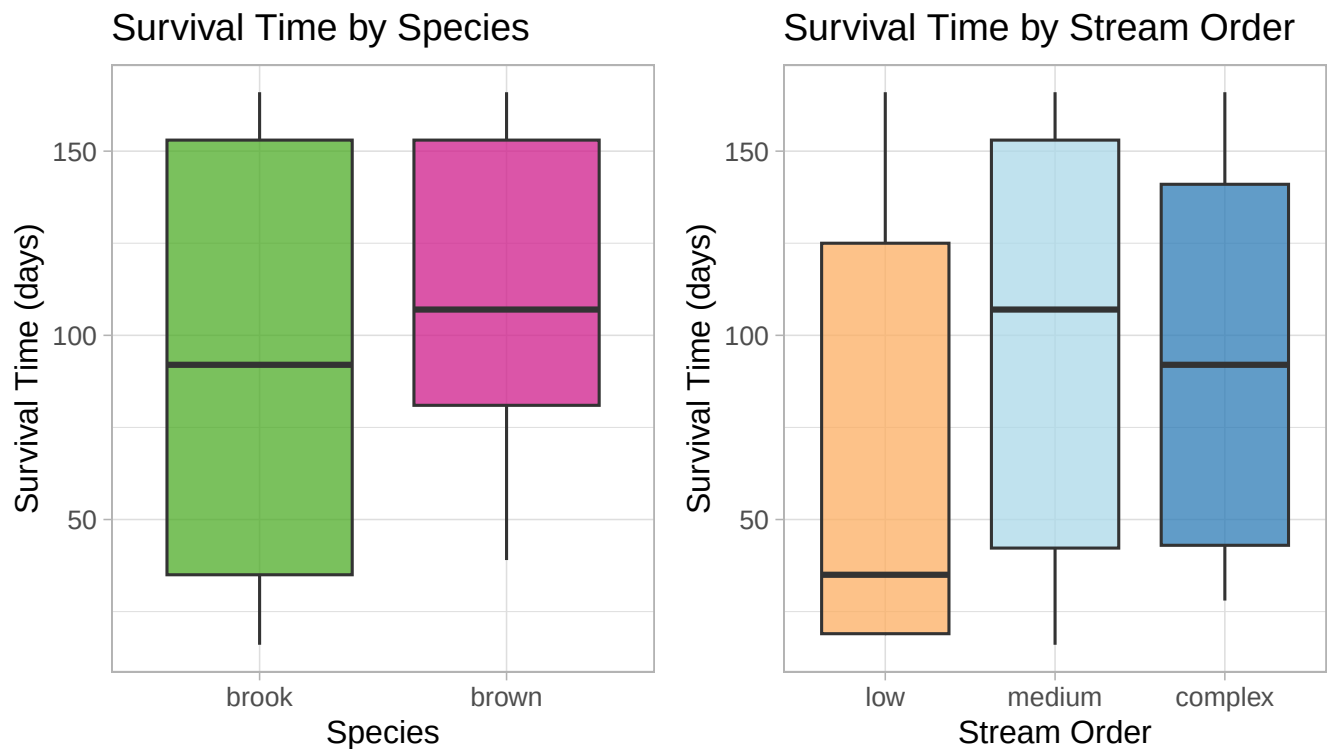
cat_species <- ggplot(trout_clean, aes(x = species, y = time2, fill = species)) +
  geom_boxplot(alpha = 0.75, outlier.shape = 16) +
  scale_fill_manual(values = c("#4dac26", "#d01c8b")) +
  labs(title = "Survival Time by Species", x = "Species", y = "Survival Time (days)") +
  theme_light(base_size = 12) + theme(legend.position = "none")

cat_stream <- ggplot(trout_clean, aes(x = stream, y = time2, fill = stream)) +
  geom_boxplot(alpha = 0.75, outlier.shape = 16) +
  scale_fill_brewer(palette = "Set2") +
  labs(title = "Survival Time by Stream", x = "Stream", y = "Survival Time (days)") +
  theme_light(base_size = 12) +
  theme(axis.text.x = element_text(angle = 30, hjust = 1), legend.position = "none")

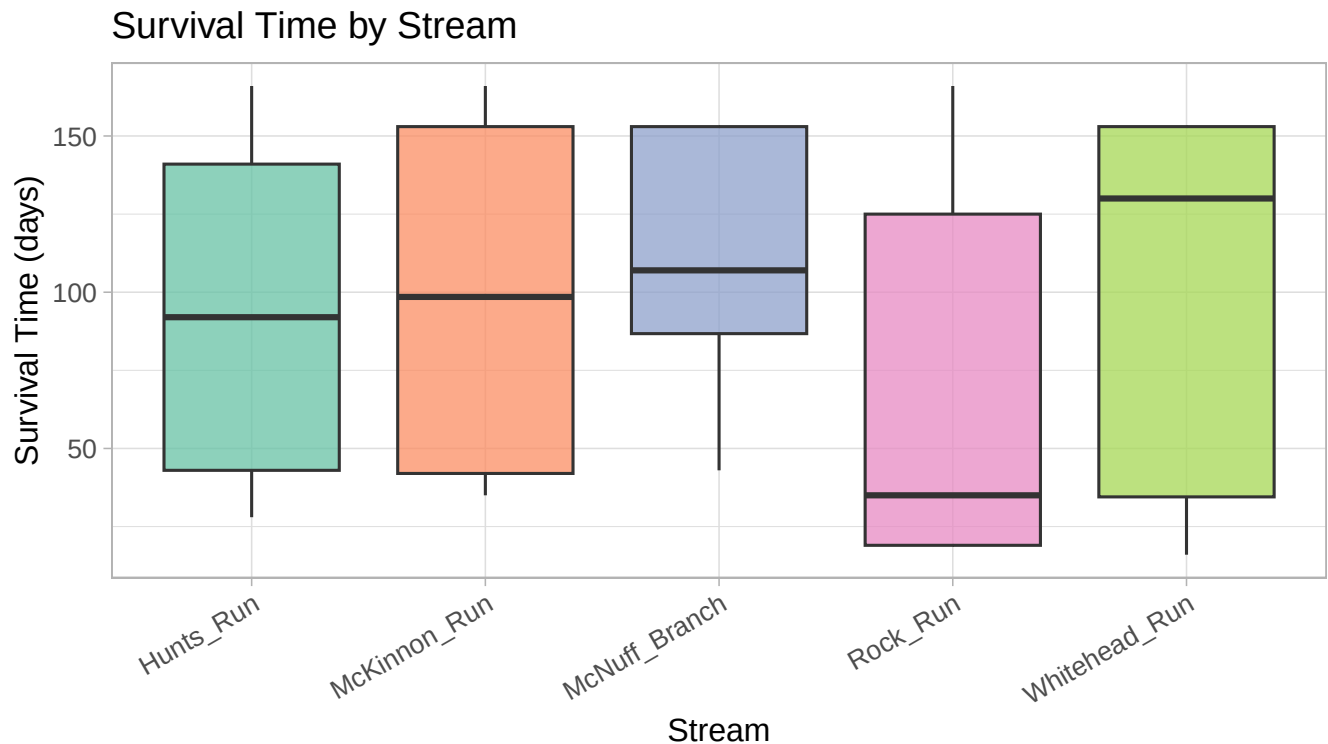
cat_order <- ggplot(trout_clean, aes(x = order, y = time2, fill = order)) +
  geom_boxplot(alpha = 0.75, outlier.shape = 16) +
  scale_fill_manual(values = c("#fdae61", "#abd9e9", "#2c7bb6")) +
  labs(title = "Survival Time by Stream Order", x = "Stream Order",
       y = "Survival Time (days)") +
  theme_light(base_size = 12) + theme(legend.position = "none")

grid.arrange(cat_species, cat_order, ncol = 2)

```



cat_stream



Brook and Brown Trout show similar median survival times with overlapping distributions. Fish in low-order streams appear to have shorter survival times, while medium-order streams show the widest variability. Differences across streams are visible but partially explained by their respective order values.

Quantitative Variables

```
summary(trout_clean[, c("Lgt", "Wgt", "cumdrain", "move_rate")])
```

	Lgt	Wgt	cumdrain	move_rate
## Min.	:161.0	Min. : 45.0	Min. : 3.809	Min. : 0.110
## 1st Qu.:	:197.8	1st Qu.: 74.0	1st Qu.:11.321	1st Qu.: 3.430
## Median :	:223.5	Median :101.0	Median :24.906	Median : 8.425
## Mean :	:234.0	Mean :141.6	Mean :35.893	Mean : 16.035
## 3rd Qu.:	:251.5	3rd Qu.:152.8	3rd Qu.:79.263	3rd Qu.: 21.047
## Max. :	:430.0	Max. :829.0	Max. :79.263	Max. :151.240

```
h1 <- ggplot(trout_clean, aes(x = Lgt)) +
  geom_histogram(bins = 18, fill = "#74add1", color = "white") +
  labs(title = "Fish Length (mm)", x = "Length (mm)", y = "Count") + theme_light()

h2 <- ggplot(trout_clean, aes(x = Wgt)) +
  geom_histogram(bins = 18, fill = "#f46d43", color = "white") +
  labs(title = "Fish Weight (g) - Raw", x = "Weight (g)", y = "Count") + theme_light()

h3 <- ggplot(trout_clean, aes(x = log_Wgt)) +
  geom_histogram(bins = 18, fill = "#fdae61", color = "white") +
  labs(title = "Fish Weight - Log Transformed", x = "log(Weight)", y = "Count") +
  theme_light()
```



```

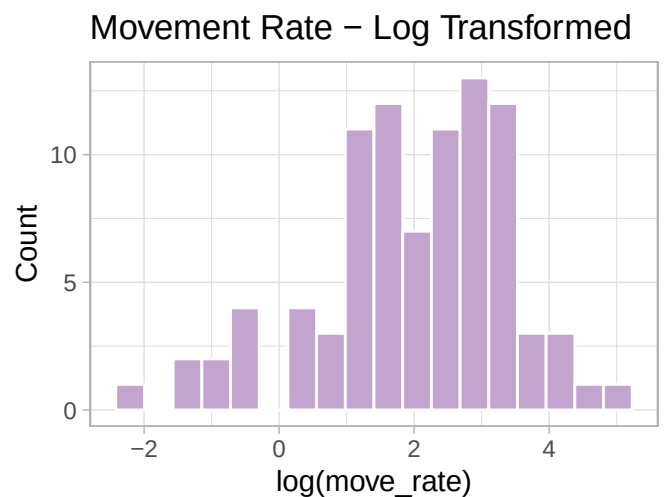
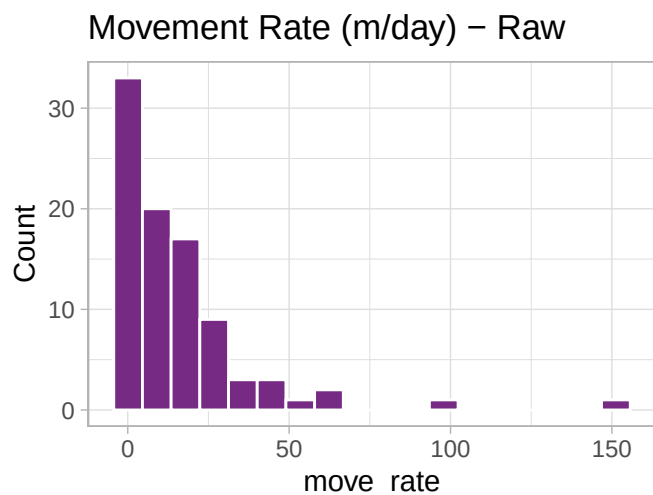
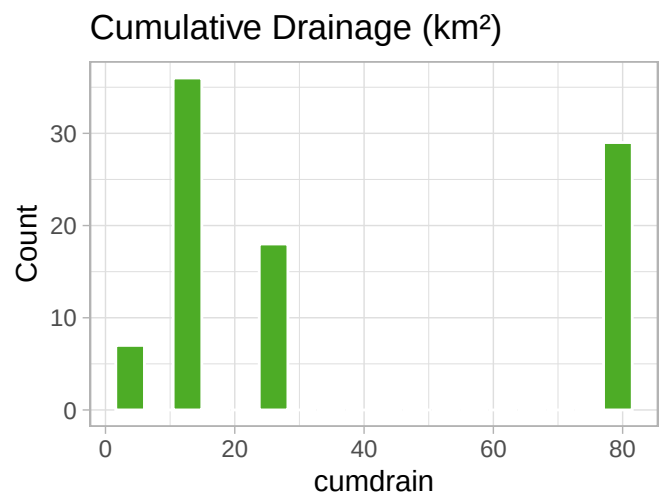
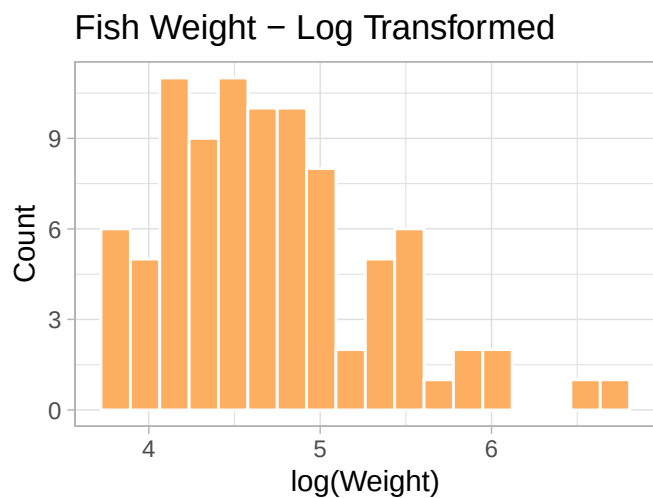
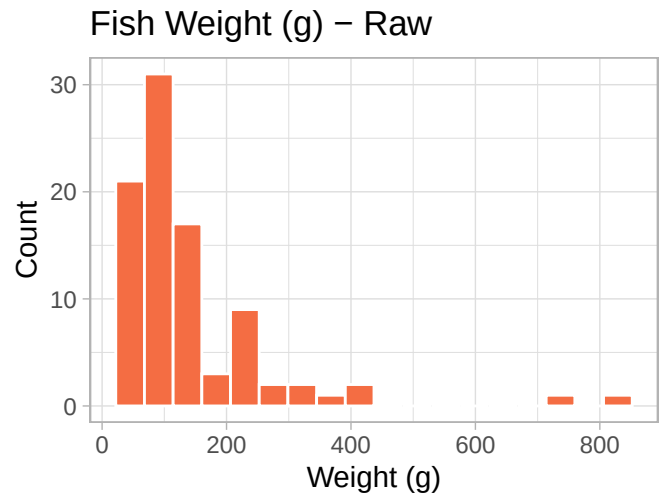
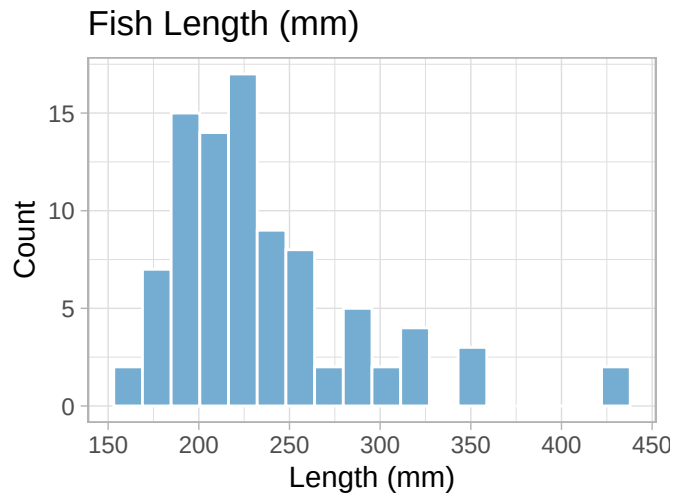
h4 <- ggplot(trout_clean, aes(x = cumdrain)) +
  geom_histogram(bins = 18, fill = "#4dac26", color = "white") +
  labs(title = "Cumulative Drainage (km2)", x = "cumdrain", y = "Count") + theme_light()

h5 <- ggplot(trout_clean, aes(x = move_rate)) +
  geom_histogram(bins = 18, fill = "#762a83", color = "white") +
  labs(title = "Movement Rate (m/day) - Raw", x = "move_rate", y = "Count") + theme_light()

h6 <- ggplot(trout_clean, aes(x = log_move_rate)) +
  geom_histogram(bins = 18, fill = "#c2a5cf", color = "white") +
  labs(title = "Movement Rate - Log Transformed", x = "log(move_rate)", y = "Count") +
  theme_light()

grid.arrange(h1, h2, h3, h4, h5, h6, ncol = 2)

```



Both `Wgt` and `move_rate` are heavily right-skewed. When we apply the log transformations there is a significant increase in symmetry and are used in Cox regression. `Lgt` is approximately normally distributed. `cumdrain` shows a bimodal shape reflecting stream-level clustering of catchment sizes.

III. Analysis

Kaplan-Meier Survival Curve & Log-Rank Test

We compared survival between Brook and Brown Trout using the Kaplan-Meier estimator and a log-rank test.

```
surv_obj <- Surv(time = trout_clean$time2, event = trout_clean$event)
km_fit <- survfit(Surv(time2, event) ~ species, data = trout_clean)
print(summary(km_fit), digits = 4)
```

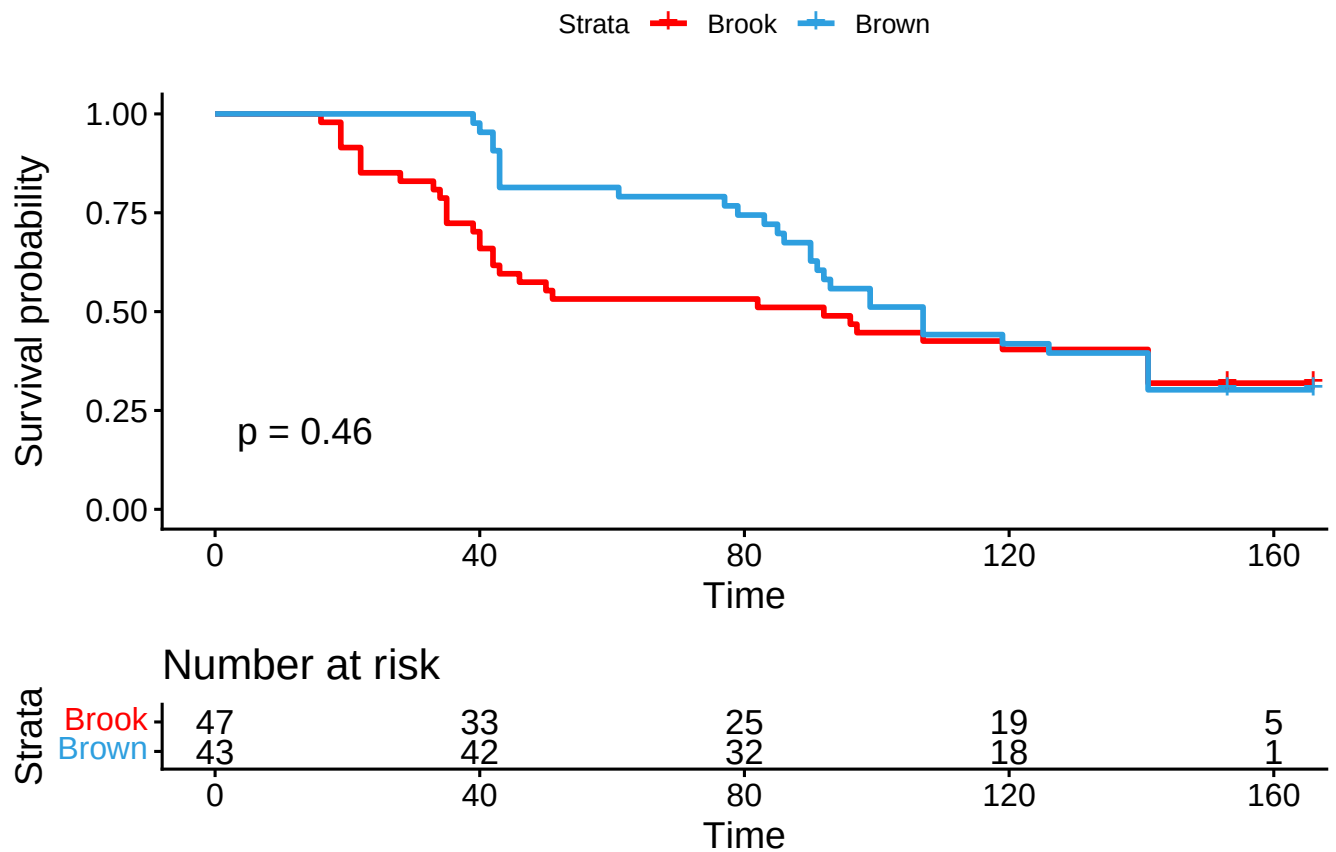


```
## Call: survfit(formula = Surv(time2, event) ~ species, data = trout_clean)
##
##               species=brook
##   time n.risk n.event survival std.err lower 95% CI upper 95% CI
##   16     47      1  0.9787 0.02105   0.9383   1.0000
##   19     46      3  0.9149 0.04070   0.8385   0.9982
##   22     43      3  0.8511 0.05193   0.7551   0.9592
##   28     40      1  0.8298 0.05482   0.7290   0.9445
##   33     39      1  0.8085 0.05739   0.7035   0.9292
##   34     38      1  0.7872 0.05970   0.6785   0.9134
##   35     37      3  0.7234 0.06525   0.6062   0.8633
##   39     34      1  0.7021 0.06671   0.5828   0.8458
##   40     33      2  0.6596 0.06912   0.5371   0.8100
##   42     31      2  0.6170 0.07091   0.4926   0.7729
##   43     29      1  0.5957 0.07158   0.4707   0.7539
##   46     28      1  0.5745 0.07212   0.4492   0.7347
##   50     27      1  0.5532 0.07252   0.4278   0.7153
##   51     26      1  0.5319 0.07278   0.4068   0.6955
##   82     25      1  0.5106 0.07292   0.3860   0.6756
##   92     24      1  0.4894 0.07292   0.3654   0.6553
##   96     23      1  0.4681 0.07278   0.3451   0.6349
##   97     22      1  0.4468 0.07252   0.3251   0.6142
##  107     21      1  0.4255 0.07212   0.3053   0.5932
##  119     20      1  0.4043 0.07158   0.2857   0.5720
##  141     19      4  0.3191 0.06799   0.2102   0.4846
##
##               species=brown
##   time n.risk n.event survival std.err lower 95% CI upper 95% CI
##   39     43      1  0.9767 0.02298   0.9327   1.0000
##   40     42      1  0.9535 0.03211   0.8926   1.0000
##   42     41      2  0.9070 0.04430   0.8242   0.9981
##   43     39      4  0.8140 0.05934   0.7056   0.9390
##   61     35      1  0.7907 0.06204   0.6780   0.9221
##   77     34      1  0.7674 0.06442   0.6510   0.9047
##   79     33      1  0.7442 0.06654   0.6246   0.8867
##   83     32      1  0.7209 0.06840   0.5986   0.8683
##   85     31      1  0.6977 0.07004   0.5731   0.8494
##   86     30      1  0.6744 0.07146   0.5479   0.8301
##   90     29      2  0.6279 0.07371   0.4988   0.7904
##   91     27      1  0.6047 0.07456   0.4748   0.7700
##   92     26      1  0.5814 0.07523   0.4512   0.7492
##   93     25      1  0.5581 0.07573   0.4278   0.7282
##   99     24      2  0.5116 0.07623   0.3821   0.6851
##  107     22      3  0.4419 0.07573   0.3158   0.6183
```

```
##    119     19      1  0.4186 0.07523      0.2943      0.5954
##    126     18      1  0.3953 0.07456      0.2732      0.5722
##    141     17      4  0.3023 0.07004      0.1920      0.4761
```

```
ggsurvplot(km_fit,
data = trout_clean,
pval = TRUE,
risk.table = TRUE,
palette = c("red", "#2E9FDF"), legend.labs = c("Brook", "Brown"), title = "Survival by Trout Species")
```

Survival by Trout Species



```
surv_diff <- survdiff(surv_obj ~ species, data = trout_clean)
surv_diff
```

```
## Call:
## survdiff(formula = surv_obj ~ species, data = trout_clean)
##
##              N Observed Expected (O-E)^2/E (O-E)^2/V
## species=brook 47      32     29.2    0.268    0.534
## species=brown 43      30     32.8    0.239    0.534
##
## Chisq= 0.5  on 1 degrees of freedom, p= 0.5
```

Hypotheses:

- H_0 : There is no difference in survival time between Brook and Brown Trout

- H_A : There is a difference in survival time between Brook and Brown Trout

Conclusion: With $p = 0.5 > \alpha = 0.05$, we fail to reject the null hypothesis. There is no statistically significant difference in survival between species based on the log-rank test alone. We proceed to Cox regression to evaluate species alongside other covariates while adjusting for potential confounders.

Cox Proportional Hazards Regression

Model Building & Backward Selection

We begin with a full model and sequentially remove variables based on collinearity findings and significance.

Model 1 — Full model (includes both stream and order):

```
surv_obj <- Surv(time = trout_clean$time2, event = trout_clean$event)
model1 <- coxph(surv_obj ~ species + Lgt + Wgt + stream + order + cumdrain + move_rate, data = trout_clean)
summary(model1)
```

```
## Call:
## coxph(formula = surv_obj ~ species + Lgt + Wgt + stream + order +
##       cumdrain + move_rate, data = trout_clean)
##
##      n= 90, number of events= 62
##
##              coef exp(coef)    se(coef)      z Pr(>|z|)
## speciesbrown    -0.2384768  0.7878269  0.3001211 -0.795   0.427
## Lgt              0.0006148  1.0006150  0.0071201  0.086   0.931
## Wgt              0.0010814  1.0010820  0.0027223  0.397   0.691
## streamMcKinnon_Run -0.0549793  0.9465047  0.3774950 -0.146   0.884
## streamMcNuff_Branch -0.2636432  0.7682476  0.3927003 -0.671   0.502
## streamRock_Run     0.3390834  1.4036604  0.5427157  0.625   0.532
## streamWhitehead_Run -0.0508912  0.9503821  0.3787845 -0.134   0.893
## ordermedium        NA         NA  0.0000000    NA     NA
## ordercomplex        NA         NA  0.0000000    NA     NA
## cumdrain           NA         NA  0.0000000    NA     NA
## move_rate          0.0017367  1.0017382  0.0060737  0.286   0.775
##
##              exp(coef) exp(-coef) lower .95 upper .95
## speciesbrown      0.7878      1.2693     0.4375     1.419
## Lgt                1.0006      0.9994     0.9867     1.015
## Wgt                1.0011      0.9989     0.9958     1.006
## streamMcKinnon_Run 0.9465      1.0565     0.4516     1.984
## streamMcNuff_Branch 0.7682      1.3017     0.3558     1.659
## streamRock_Run     1.4037      0.7124     0.4845     4.067
## streamWhitehead_Run 0.9504      1.0522     0.4524     1.997
## ordermedium        NA         NA         NA         NA
## ordercomplex        NA         NA         NA         NA
## cumdrain           NA         NA         NA         NA
## move_rate          1.0017      0.9983     0.9899     1.014
##
## Concordance= 0.587 (se = 0.035 )
## Likelihood ratio test= 3.73 on 8 df,  p=0.9
## Wald test              = 3.89 on 8 df,  p=0.9
## Score (logrank) test = 3.99 on 8 df,  p=0.9
```

```
table(trout_clean$order, trout_clean$stream)
```

```
##
##           Hunts_Run McKinnon_Run McNuff_Branch Rock_Run Whitehead_Run
## low              0              0              0          7              0
## medium           0              18             18          0             18
## complex          29              0              0          0              0
```

stream and order are perfectly collinear —each stream belongs to exactly one order level — causing rank-deficiency. One must be dropped.

Model 2 — Drop order, keep stream:

```
model2 <- coxph(surv_obj ~ species + Lgt + Wgt + stream +
  cumdrain + move_rate , data = trout_clean)
summary(model2)
```

```
## Call:
## coxph(formula = surv_obj ~ species + Lgt + Wgt + stream + cumdrain +
##   move_rate, data = trout_clean)
##
##   n= 90, number of events= 62
##
##               coef exp(coef) se(coef)      z Pr(>|z|)
## speciesbrown   -0.2384768  0.7878269  0.3001211 -0.795    0.427
## Lgt             0.0006148  1.0006150  0.0071201  0.086    0.931
## Wgt            0.0010814  1.0010820  0.0027223  0.397    0.691
## streamMcKinnon_Run -0.0549793  0.9465047  0.3774950 -0.146    0.884
## streamMcNuff_Branch -0.2636432  0.7682476  0.3927003 -0.671    0.502
## streamRock_Run    0.3390834  1.4036604  0.5427157  0.625    0.532
## streamWhitehead_Run -0.0508912  0.9503821  0.3787845 -0.134    0.893
## cumdrain         NA         NA  0.0000000    NA      NA
## move_rate         0.0017367  1.0017382  0.0060737  0.286    0.775
##
##               exp(coef) exp(-coef) lower .95 upper .95
## speciesbrown    0.7878    1.2693    0.4375    1.419
## Lgt             1.0006    0.9994    0.9867    1.015
## Wgt            1.0011    0.9989    0.9958    1.006
## streamMcKinnon_Run 0.9465    1.0565    0.4516    1.984
## streamMcNuff_Branch 0.7682    1.3017    0.3558    1.659
## streamRock_Run    1.4037    0.7124    0.4845    4.067
## streamWhitehead_Run 0.9504    1.0522    0.4524    1.997
## cumdrain         NA         NA      NA      NA
## move_rate        1.0017    0.9983    0.9899    1.014
##
## Concordance= 0.587 (se = 0.035 )
## Likelihood ratio test= 3.73 on 8 df,  p=0.9
## Wald test              = 3.89 on 8 df,  p=0.9
## Score (logrank) test = 3.99 on 8 df,  p=0.9
```

Model 2 produces NA estimates for cumdrain due to additional collinearity between cumdrain and the stream dummy variables. This model is not usable.

Model 3 — Drop stream, keep order; use log-transformed predictors; exclude Lgt:

```
model3 <- coxph(surv_obj ~ species + log_Wgt + order +
  cumdrain + log_move_rate , data = trout_clean)
```

```
summary(model3)
```

```
## Call:
## coxph(formula = surv_obj ~ species + log_Wgt + order + cumdrain +
##       log_move_rate, data = trout_clean)
##
##      n= 90, number of events= 62
##
##              coef exp(coef) se(coef)      z Pr(>|z|)
## speciesbrown  -0.35727   0.69959  0.29550 -1.209  0.2267
## log_Wgt        0.21039   1.23416  0.23919  0.880  0.3791
## ordermedium   -0.58753   0.55570  0.62639 -0.938  0.3483
## ordercomplex  -2.17837   0.11323  2.34778 -0.928  0.3535
## cumdrain       0.02930   1.02974  0.03063  0.957  0.3388
## log_move_rate -0.23168   0.79320  0.10799 -2.145  0.0319 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##              exp(coef) exp(-coef) lower .95 upper .95
## speciesbrown    0.6996     1.4294  0.392019    1.2485
## log_Wgt          1.2342     0.8103  0.772280    1.9723
## ordermedium      0.5557     1.7995  0.162798    1.8968
## ordercomplex     0.1132     8.8319  0.001136   11.2817
## cumdrain         1.0297     0.9711  0.969729    1.0935
## log_move_rate    0.7932     1.2607  0.641898    0.9802
##
## Concordance= 0.6 (se = 0.039 )
## Likelihood ratio test= 6.65 on 6 df,  p=0.4
## Wald test              = 7.23 on 6 df,  p=0.3
## Score (logrank) test = 7.28 on 6 df,  p=0.3
```

Model Choice

```
AIC(model1, model2, model3)
```

```
##      df      AIC
## model1  8 512.7971
## model2  8 512.7971
## model3  6 505.8723
```

Model 3 resolves all collinearity issues and is our **final model**. `log_Wgt` replaces raw `Wgt` for better linearity on the log-hazard scale, `Lgt` is excluded due to its near-perfect correlation with weight ($r = 0.90$), and `log_move_rate` is used given the right skew of the raw movement variable.

Investigating the AIC confirms **Model 3** as the correct choice with 6 parameters (AIC = 505.8723).

IV. Key Findings

Among all predictors in our final model (Model 3), only movement rate was statistically significant (HR = 0.793, $p = 0.032$), indicating that increased movement is associated with reduced hazard and improved survival. Other variables including; `cumdrain`, `log_Wgt`, `order`, and `species` were insignificant or too noisy to reliably predict the effects seen in the data. The overall model was not statistically significant (Likelihood ratio test $p = 0.40$), and concordance was moderate (0.60), suggesting moderate predictive power.

Hazard Ratio Interpretation

Fish with higher movement rates have significantly lower hazard, suggesting increased survival. For a one-unit increase in log movement rate, the hazard decreases by about 21% ($1 - 0.793$), given all other variables are held constant. As this is the only significant variable it can be viewed as a reliable estimate. This also may suggest that more active fish are better able to avoid predators or occupy more favorable habitats, improving their survival probability.

Brown trout were observed to have a Hazard Ratio (0.70), which can be interpreted as having a 30% lower hazard rate than Brook trout (the reference species). Though in the model this variable wasn't significant so no real effect can be concluded.

The `order` variable was also insignificant in the model with $HR = 0.56$ (Medium), and $HR = 0.11$ (Complex), This is a very large confidence interval, therefore any interpretations would be unreliable.

The variable `cumdrain` was also statistically insignificant with a hazard ratio (1.03), which means the amount of land drainage into the stream has no effect on survival.

V. Model Assumptions

Proportional Hazards

- H_0 : The log hazard ratio for each predictor is constant over time.
- H_a : The log hazard ratio changes over time.

##		chisq	df	p
##	species	11.0305	1	0.0009
##	log_Wgt	0.0614	1	0.8043
##	order	8.2545	2	0.0161
##	cumdrain	4.0428	1	0.0444
##	log_move_rate	10.0887	1	0.0015
##	GLOBAL	28.8321	6	6.5e-05

The proportional hazards assumption is violated for most predictors in Model 3. The variables `species`, `order`, `cumulative drainage (cumdrain)`, and `log move rate` all have p-values below .05, indicating we reject H_0 for each predictor and conclude that their log hazard ratios are not constant over time.

Proportional Hazards ~ Correction Since effects in our dataset tend to change overtime such as `move_rate` and `cumdrain`, future studies should include time covariates that include the interaction between time and the predictors in our model. This could better model the influence these variables have on survival time over the study period. Furthermore, since the proportional hazards assumption is violated by categorical variables in our model, a stratified Cox model is a correction we could try to explore in future work. Rather than assuming a constant hazard ratio over time, a stratified model allows each group, for example species or stream order, to have its own baseline hazard function.

Linearity

```
Y <- resid(model3, type = "martingale")
x1 <- trout_clean$log_Wgt
x2 <- trout_clean$cumdrain
x3 <- trout_clean$log_move_rate

par(mfrow = c(3,1))

# Log Weight
plot(Y~x1, pch = 20, col = "darkgrey",
```



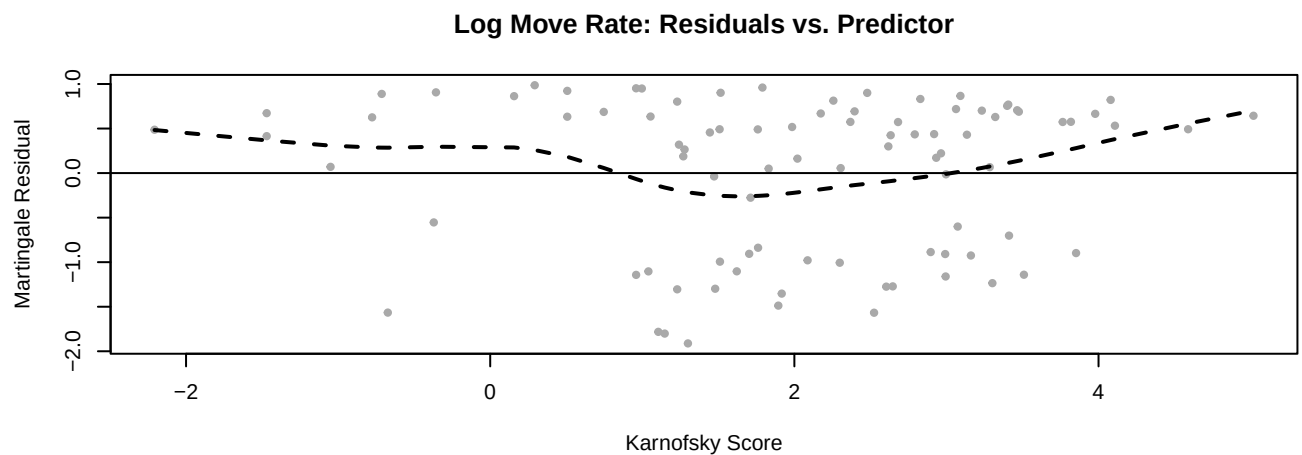
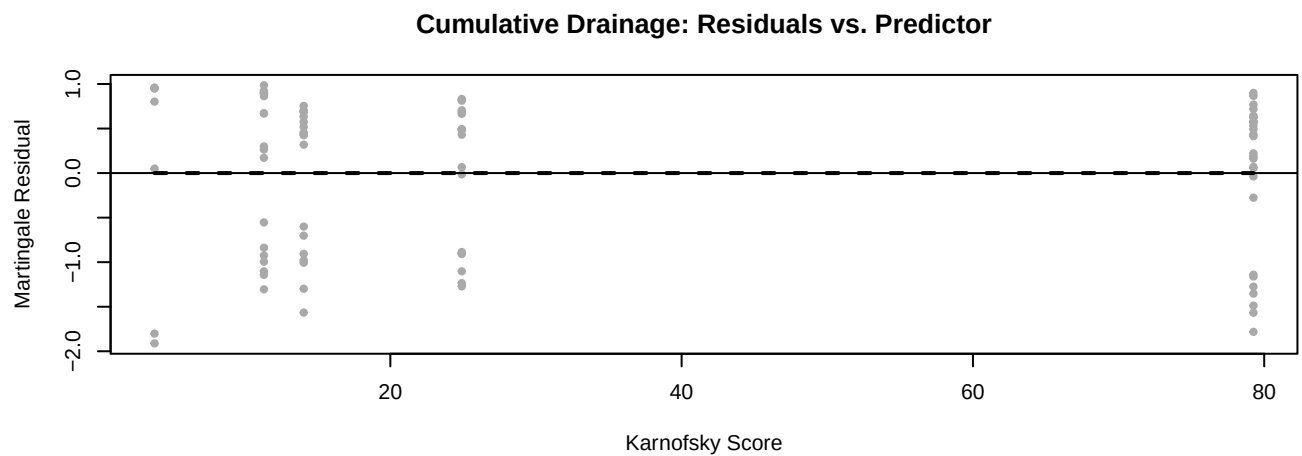
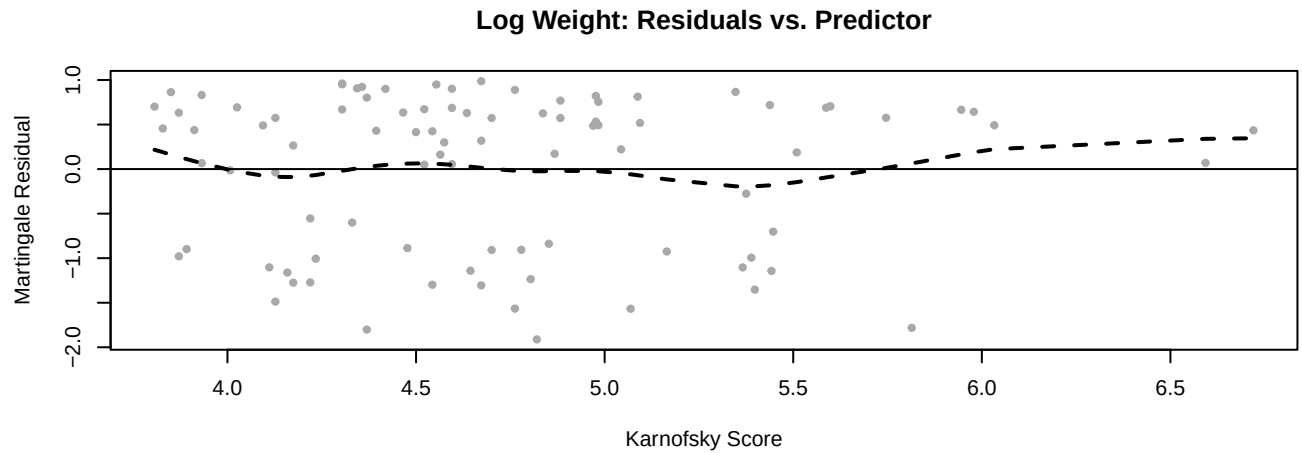
```

    ylab = "Martingale Residual",
    xlab = "Karnofsky Score",
    main = "Log Weight: Residuals vs. Predictor")
abline(h = 0)
lines(smooth.spline(Y~x1, df = 6), lty =2, lwd =2)

# Cumdrain
plot(Y~x2, pch = 20, col = "darkgrey",
     ylab = "Martingale Residual",
     xlab = "Karnofsky Score",
     main = "Cumulative Drainage: Residuals vs. Predictor")
abline(h = 0)
lines(smooth.spline(Y~x2, df = 6), lty =2, lwd =2)

# Log Move Rate
plot(Y~x3, pch = 20, col = "darkgrey",
     ylab = "Martingale Residual",
     xlab = "Karnofsky Score",
     main = "Log Move Rate: Residuals vs. Predictor")
abline(h = 0)
lines(smooth.spline(Y~x3, df = 6), lty =2, lwd =2)

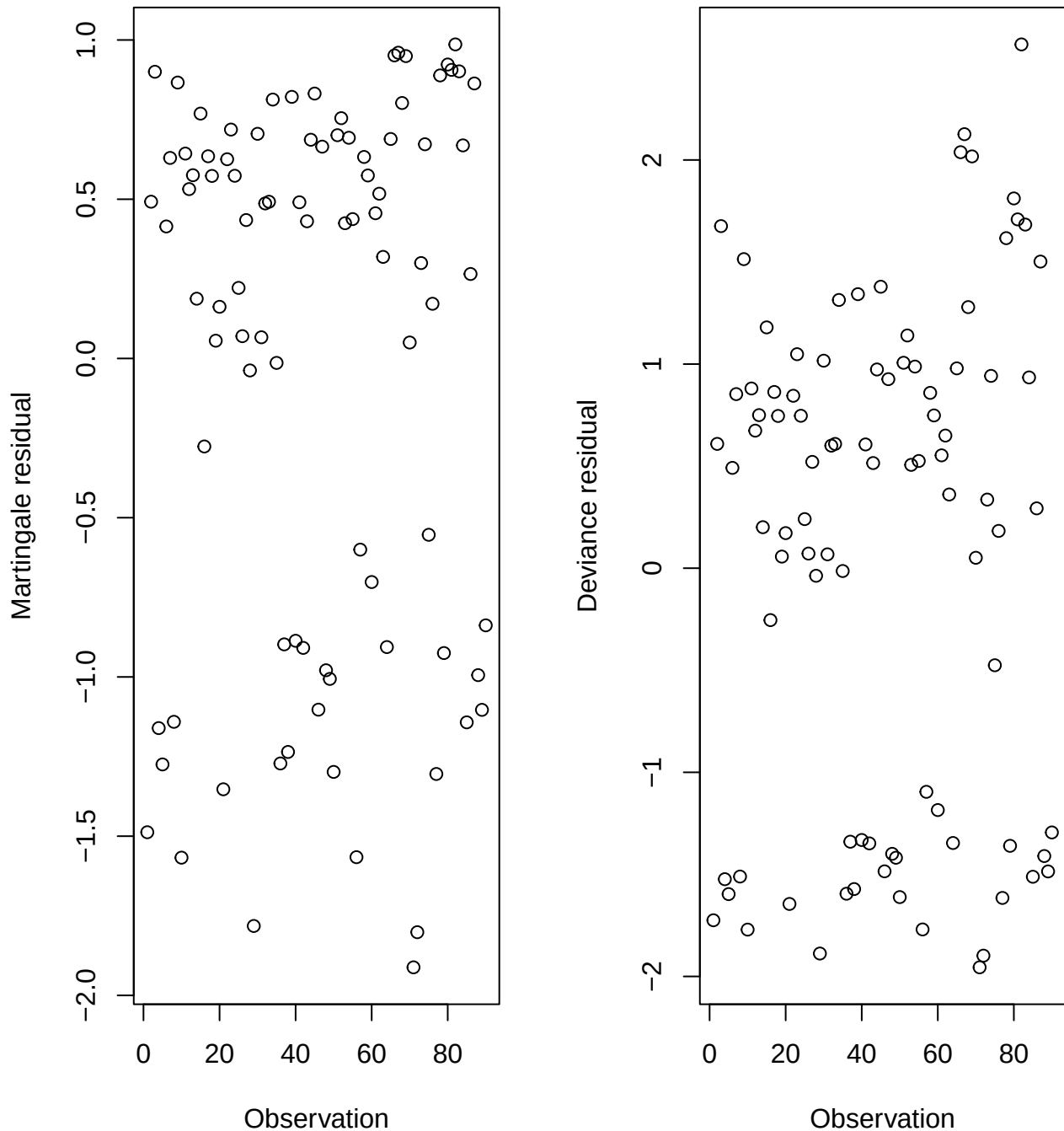
```



To assess linearity, Martingale residuals from Model 3 were plotted against each continuous predictor. For $\log(\text{weight})$, the smoothed spline remains fairly close to 0, suggesting the linearity assumption is reasonably satisfied for this variable. The Cumulative Drainage remains flat at zero, indicating a linear relationship between cumdrain and log-hazard and, furthermore, that the linearity assumption is satisfied. Lastly, for $\log(\text{move rate})$, the spline shows a slight curve, indicative of a mild departure from linearity. Since this deviation is not extreme and the log transformation has already been applied, we will consider this approximately satisfied.

Outliers

```
par(mfrow=c(1,2))
plot(resid(model3, type = "martingale"),
     ylab = "Martingale residual", xlab = "Observation")
plot(resid(model3, type = "deviance"),
     ylab = "Deviance residual", xlab = "Observation")
```



The Martingale residuals range from about -2.0 to 1.0, and a few observations fall below -1.5. However there is not a singular observation that appears to be considered an extreme outlier. In the deviance residuals plot, most observations fall between -2 and 2. One observation falls outside of this and has a value of about

3.0. This is worth noting, but overall there no extreme outliers that would harm the model estimates.

Independence

Since each fish is an individually tagged animal, this assumption is satisfied.

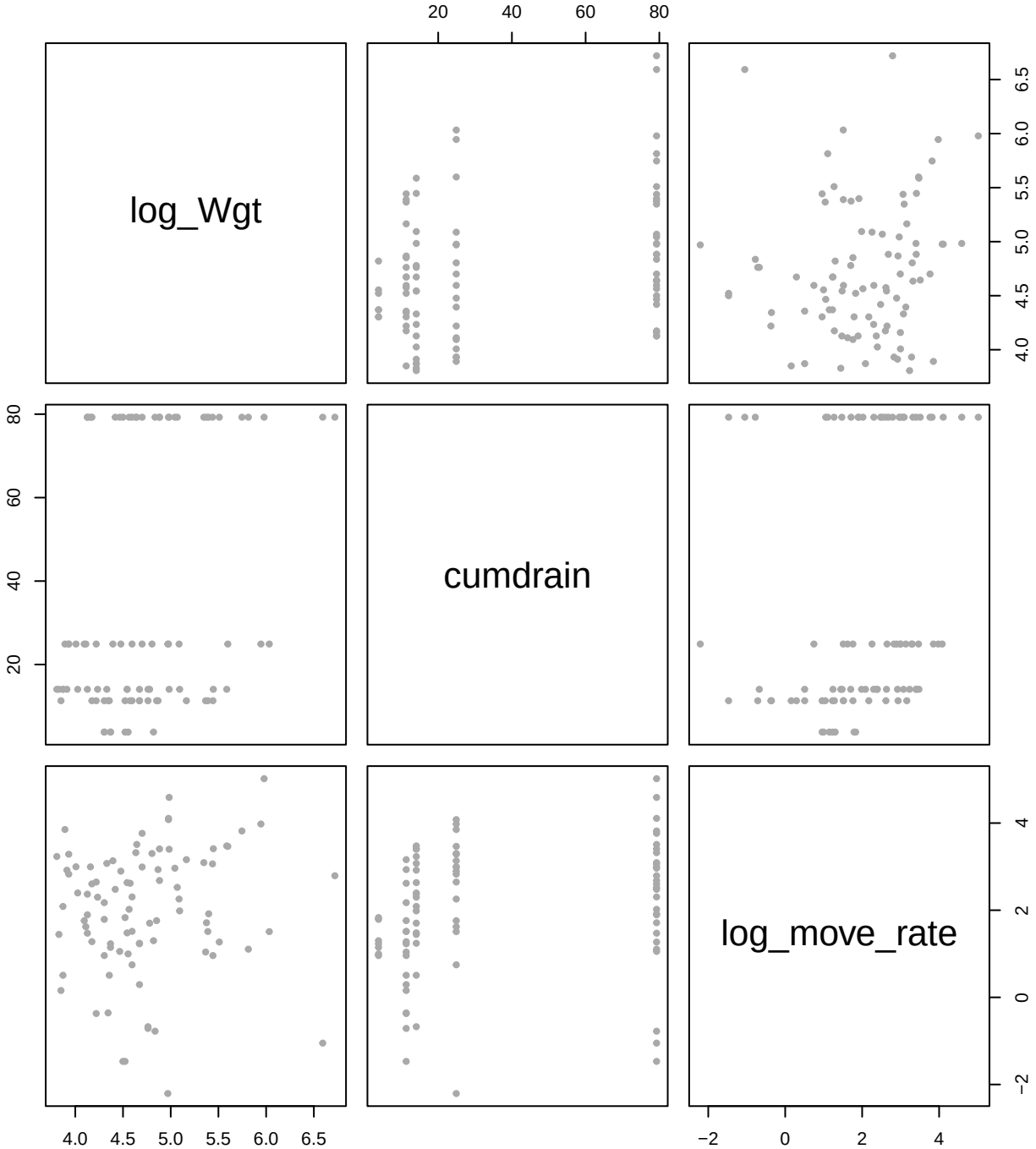
Multicollinearity

```
# Correlation matrix
cont_vars <- trout_clean[,c("log_Wgt", "cumdrain", "log_move_rate")]
cor(cont_vars)

##           log_Wgt  cumdrain log_move_rate
## log_Wgt      1.0000000 0.3389213    0.1019370
## cumdrain      0.3389213 1.0000000    0.2213707
## log_move_rate 0.1019370 0.2213707    1.0000000

# Scatter plot
pairs(trout_clean[,c("log_Wgt", "cumdrain", "log_move_rate")],
      main = "Scatterplot Matrix of Continuous Predictors",
      pch = 20, col = "darkgrey")
```

Scatterplot Matrix of Continuous Predictors



The correlation matrix indicates low pairwise correlations among all three continuous predictors of the final model. Log Weight and cumulative drainage have the highest correlations of about .34. The scatter plot matrix confirms this, as no pair of predictors depicts a strong trend. There is no evidence of multicollinearity among the predictors.

VI. Limitations

Several limitations should be considered when interpreting the results. The first is a violation of the proportional hazards assumption for most predictors in our final model. This indicates that the effect of species, order, cumulative drainage, and log movement rate on survival likely changes over time. This violates a core assumption of Cox regression. Secondly, nine fish were excluded because their movement rate data were missing. Unfortunately, if this missingness is related to fish condition or mortality risk, the sample may not be fully representative, potentially introducing selection bias. Third, while order was retained across streams, aggregating across five streams may obscure stream-specific effects on survival. Taken together, the overall model fit was not statistically significant (likelihood ratio chi-square statistic = 6.65 on 6 df, $p = .4$), suggesting the model has limited predictive ability. The sample size ($n = 90$; 62 events) is moderate for ecological survival analysis and provides sufficient events for model estimation. However, the inclusion of multiple predictors and inherent ecological variability may reduce statistical power to detect weaker effects.

VII. Conclusion

This study sought to explore the survival rate of the trout over time, comparing them at the stream level. Using Kaplan-Meier estimation and Cox proportional hazard regression, we analyzed about 90 Brook and Brown Trout across 5 Pennsylvania streams over a 153-day period. The Kaplan-Meier analysis found no statistically significant difference in survival between Brook and Brown trout across the groups. The log-rank test found that the difference between species was also not significant. This suggests that the two species have similar survival trajectories within the ecological conditions of the study.

In the Cox regression model, log movement rate was the only statistically significant predictor for survival, indicating that more active fish show a reduced hazard of death. This reflects the ability of higher moving fish to access better habitats, avoid predation, and increase overall longevity. All other predictors considered - species, weight, stream order, and cumulative drainage were not statistically significant.

There are several limitations to the study, described in greater detail in section 6, but primarily, the model's statistical significance is of concern, given that the overall likelihood ratio test was not significant. Future research should include more fish to provide a more in-depth analysis, particularly in underrepresented stream order categories. Furthermore, time-varying coefficient models should be employed to address the identified proportional hazards in this study in future studies.

VIII. Sources

- “Department of the Interior - Brook and Brown Trout Survival in a North-Central Pennsylvania Watershed.” Data.gov, U.S. Fish and Wildlife Service, 2023, catalog.data.gov/dataset/brook-and-brown-trout-survival-in-a-north-central-pennsylvania-watershed?from_hint=eyJxIjoic3VydmI2YWwgIiwic29ydCI6InJlbGV2YW5jZ. Accessed 29 Apr. 2026.